



Realtime Diagnostic Report Fraud Detection

Protecting Patient Health with Event-Driven Agentic AI



Confluent Kafka



Powered By



Apache Flink



watsonx Orchestrate

The Problem: A Patient Safety Crisis



Fake Diagnostic Reports Can Be Life-Threatening

Millions of patients receive fraudulent lab results leading to misdiagnosis, delayed treatment, and preventable deaths



Missed Critical Conditions

Cancer, diabetes, heart disease go undetected until too late



Delayed Treatment

False 'normal' results waste precious time while conditions worsen



Wrong Medications

Incorrect diagnoses lead to harmful or ineffective treatments

Root Cause: Two Major Fraud Patterns



Sink Test Fraud

Labs showing >85% 'all normal' reports — samples discarded without testing



Kickback Fraud

Doctors with >80% referrals to single lab — commission-based unnecessary tests

The Solution: Agentic AI for Patient Protection



Patient-Facing AI Agent with Real-Time Fraud Detection

Built on watsonx Orchestrate, powered by streaming data from Confluent Kafka and Apache Flink — delivering instant fraud alerts

How Patients Use the Agent:



"Is Quick Lab Diagnostics in Delhi trustworthy?"

HEALTH SAFETY ALERT

FRAUD PATTERN: Sink Test Fraud
Normal Reports: 100% (Threshold: >85%)
 Suspicious patterns detected.
→ Get second opinion from NABL lab

Agent Capabilities:

Lab Pattern Check

Analyzes lab's normal report ratio

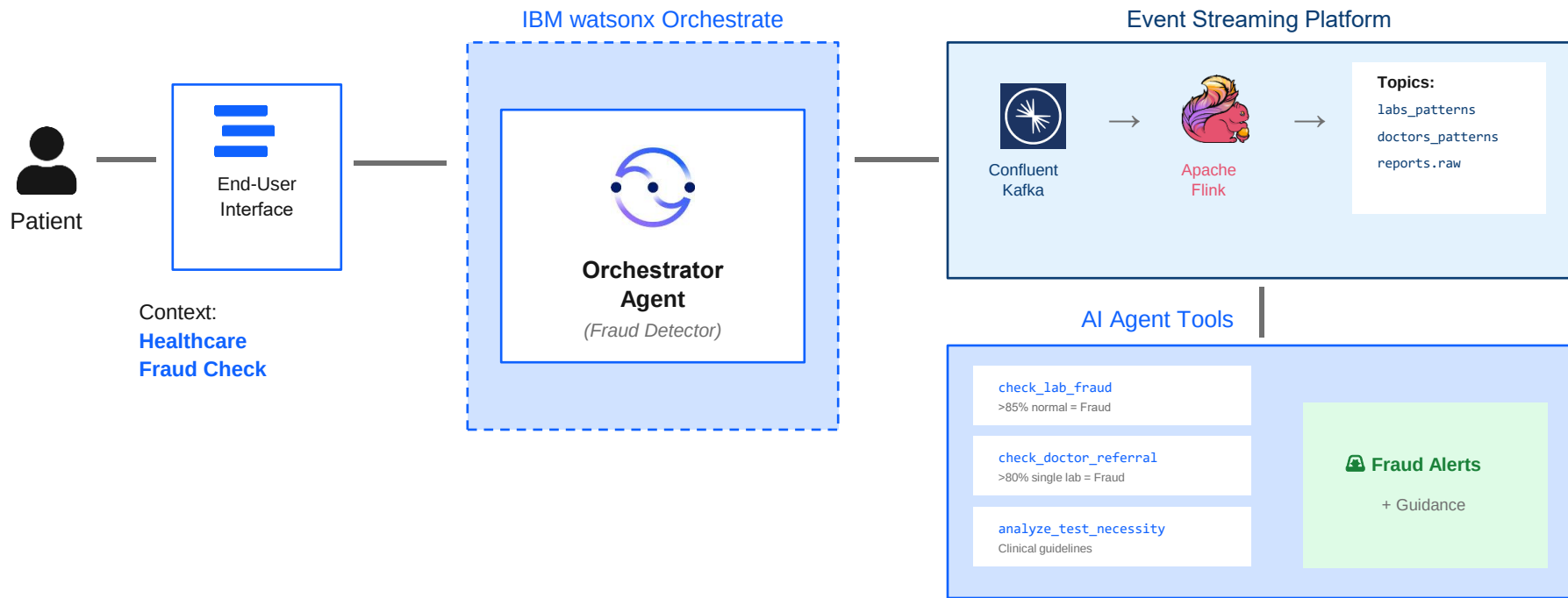
Doctor Referral Check

Detects suspicious referral patterns

Test Necessity Check

Validates if tests match symptoms

Solution Architecture



Data Flow: Patient Query → Orchestrator Agent → Kafka Topics + Flink Processing → AI Tools Analysis → Fraud Alert to Patient

Technical Components



Confluent Kafka

- Event streaming platform
- Avro schema serialization
- Schema Registry
- Topics: labs_patterns, doctors_patterns



Apache Flink

- Stream processing engine
- Real-time pattern detection
- Aggregates fraud metrics
- Sub-second latency



watsonx Orchestrate

- AI agent framework
- Python tool integration
- Knowledge base
- GPT-OSS-120b model

Python Tools (watsonx Orchestrate Agent)

`check_lab_fraud_pattern.py`

Queries Kafka for lab patterns, detects >85% normal

>85% = Fraud

`check_doctor_referral_pattern.py`

Analyzes doctor-lab referral concentration

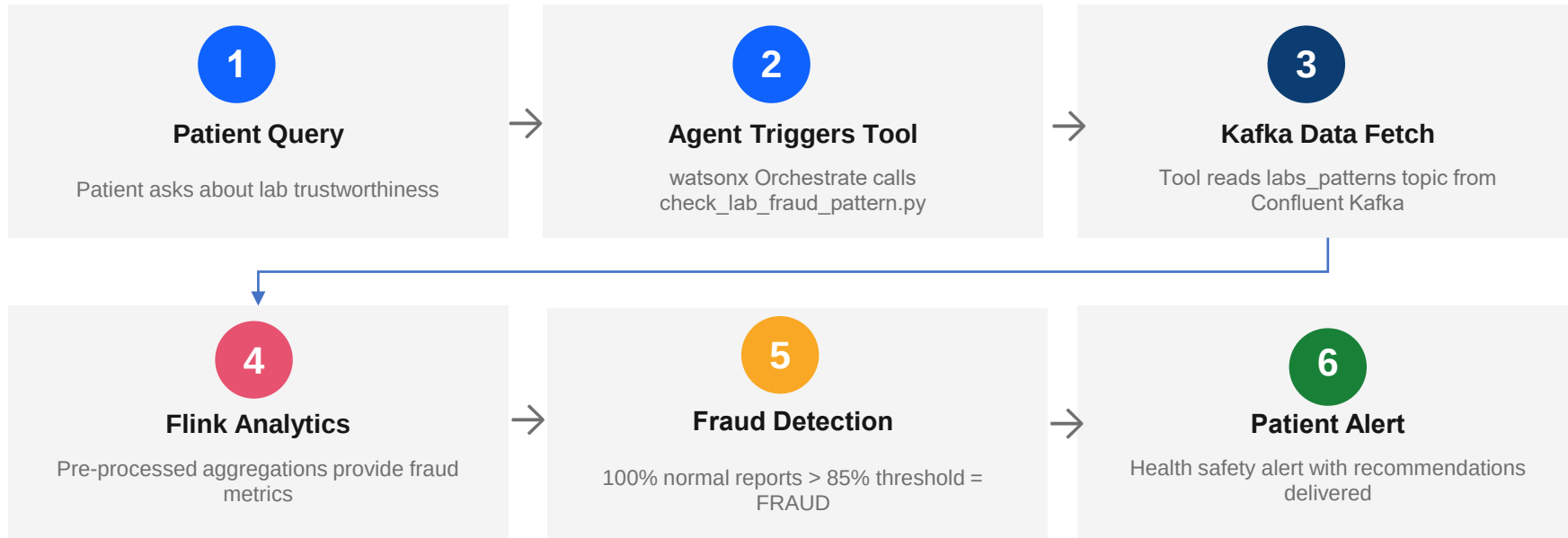
>80% = Fraud

`analyze_test_necessity.py`

Validates ordered tests against medical guidelines

Clinical Rules

How It Works: Live Demo Flow



Sample Agent Response:

 **FRAUD ALERT:** Quick Lab Diagnostics shows 100% normal reports (threshold >85%). Pattern: Sink Test Fraud. Action: Get second opinion from NABL-accredited lab.

Benefits & Patient Impact



Protecting Patient Health — The Primary Goal

Preventing misdiagnosis, delayed treatment, and unnecessary medical interventions through real-time fraud detection

Real-Time

Instant detection

<30 sec

Response time

85% / 80%

Detection accuracy

For Patients

- ✓ Avoid fake test results
- ✓ Get accurate diagnoses
- ✓ Make informed decisions
- ✓ Prevent wrong treatments

For Healthcare System

- ✓ Identify fraudulent labs
- ✓ Track suspicious referrals
- ✓ Improve industry trust
- ✓ Reduce malpractice risks

Financial Impact

- ✓ Save ₹2,000-3,000 per case
- ✓ Avoid unnecessary retests
- ✓ Prevent wrong treatment costs
- ✓ Reduce insurance fraud



Protecting Patient Health

with Event-Driven Agentic AI



Real-time fraud detection preventing patient harm from fake lab reports



Event-driven: Confluent Kafka → Apache Flink → watsonx Orchestrate



Detects Sink Test (>85%) and Kickback (>80%) fraud patterns instantly



Empowers patients with actionable health safety alerts and guidance

Questions?